Cyanosis can be caused by small quantities of circulating methemoglobin (Hb Fe^{3+}) and by even smaller quantities of sulfhemoglobin (Chap. 103); both of these hemoglobin derivatives impair oxygen delivery to the tissues. Although they are uncommon causes of cyanosis, these abnormal hemoglobin species should be sought by spectroscopy when cyanosis is not readily explained by malfunction of the circulatory or respiratory systems. Generally, digital clubbing does not occur with these conditions.

Peripheral Cyanosis Probably the most common cause of peripheral cyanosis is the normal vasoconstriction resulting from exposure to cold air or water. When cardiac output is reduced, cutaneous vasoconstriction occurs as a compensatory mechanism so that blood is diverted from the skin to more vital areas such as the CNS and heart, and cyanosis of the extremities may result even though the arterial blood is normally saturated.

Arterial obstruction to an extremity, as with an embolus, or arteriolar constriction, as in cold-induced vasospasm (Raynaud's phenomenon) (Chap. 292), generally results in pallor and coldness, and there may be associated cyanosis. Venous obstruction, as in thrombophlebitis or deep venous thrombosis, dilates the subpapillary venous plexuses and thereby intensifies cyanosis.

APPROACH TO THE PATIENT

Cyanosis

Certain features are important in arriving at the cause of cyanosis:

1. It is important to ascertain the time of onset of cyanosis. Cyanosis present since birth or infancy is usually due to congenital heart disease.
2. Central and peripheral cyanosis must be differentiated. Evidence of disorders of the respiratory or cardiovascular systems is helpful. Massage or gentle warming of a cyanotic extremity will increase peripheral blood flow and abolish peripheral, but not central, cyanosis.
3. The presence or absence of clubbing of the digits (see below) should be ascertained. The combination of cyanosis and clubbing is frequent in patients with congenital heart disease and right-to-left shunting and is seen occasionally in patients with pulmonary disease, such as lung abscess or pulmonary arteriovenous fistulae. In contrast, peripheral cyanosis or acutely developing central cyanosis is *not* associated with clubbed digits.
4. Pa_{O_2} and Sa_{O_2} should be determined, and in patients with cyanosis in whom the mechanism is obscure, spectroscopic examination of the blood should be performed to look for abnormal types of hemoglobin (critical in the differential diagnosis of cyanosis).

CLUBBING

The selective bulbous enlargement of the distal segments of the fingers and toes due to proliferation of connective tissue, particularly on the dorsal surface, is termed *clubbing*; there is also increased sponginess of the soft tissue at the base of the clubbed nail. Clubbing may be hereditary, idiopathic, or acquired and associated with a variety of disorders, including cyanotic congenital heart disease (see above), infective endocarditis, and a variety of pulmonary conditions (among them primary and metastatic lung cancer, bronchiectasis, asbestosis, sarcoidosis, lung abscess, cystic fibrosis, tuberculosis, and mesothelioma), as well as with some gastrointestinal diseases (including inflammatory bowel disease and hepatic cirrhosis). In some instances, it is occupational, for example, in jackhammer operators.

Clubbing in patients with primary and metastatic lung cancer, mesothelioma, bronchiectasis, or hepatic cirrhosis may be associated with *hypertrophic osteoarthropathy*. In this condition, the subperiosteal formation of new bone in the distal diaphyses of the long bones of the extremities causes pain and symmetric arthritis-like changes in the shoulders, knees, ankles, wrists, and elbows. The diagnosis of hypertrophic osteoarthropathy may be confirmed by bone radiograph or magnetic resonance imaging (MRI). Although the mechanism of clubbing is unclear, it appears to be secondary to humoral substances that cause dilation of the vessels of the distal digits as well as growth factors released from platelet precursors in the digital circulation. In certain circumstances, clubbing is reversible, such as following lung transplantation for cystic fibrosis.

FURTHER READING

Essouma M et al: Epidemiology of digital clubbing and hypertrophic osteoarthropathy: A systematic review and meta-analysis. J Clin Rheumatol 28:104, 2022.

Gao H et al: Acquired methemoglobinemia: A systematic review of reported cases. Transfus Apher Sci 61:103299, 2022.

MacIntyre NR: Tissue hypoxia: Implications for the respiratory clinician. Respir Care 59:1590, 2014.

43 Edema

Joseph Loscalzo

PLASMA AND INTERSTITIAL FLUID EXCHANGE

Approximately two-thirds of total body water is intracellular and one-third is extracellular. One-fourth of the latter is in the plasma, and the remainder comprises the interstitial fluid. Edema represents an excess of interstitial fluid that has become evident clinically.

There is constant interchange of fluid between the two compartments of the extracellular fluid. The hydrostatic pressure within the capillaries and the colloid oncotic pressure in the interstitial fluid promote the movement of water and diffusible solutes from plasma to the interstitium. This movement is most prominent at the arterial origin of the capillary and falls progressively with the decline in intracapillary pressure and the rise in oncotic pressure toward the venular end. Fluid is returned from the interstitial space into the vascular system largely through the lymphatic system. These interchanges of fluids are normally balanced so that the volumes of the intravascular and interstitial compartments remain constant. However, a net movement of fluid from the intravascular to the interstitial spaces takes place and may be responsible for the development of edema under the following conditions: (1) an increase in intracapillary hydrostatic pressure; (2) inadequate lymphatic drainage; (3) reductions in the oncotic pressure of plasma; (4) damage to or dysfunction of the capillary endothelial barrier; and (5) increases in the oncotic pressure of the interstitial space.

REDUCTION OF EFFECTIVE ARTERIAL VOLUME

In many forms of edema, the effective arterial blood volume, a parameter that represents the filling of the arterial tree and that effectively perfuses the tissues, is reduced. Underfilling of the arterial tree may be caused by a reduction of cardiac output and/or systemic vascular resistance, by the pooling of blood in the splanchnic veins (as in cirrhosis), and by hypoalbuminemia (Fig. 43-1*A*). As a consequence of this underfilling, a series of physiological responses designed to restore the effective arterial volume to normal are set into motion. A key element of these responses is the renal retention of sodium and, therefore, water, thereby restoring effective arterial volume, but sometimes also leading to the development or intensification of edema.

RENAL FACTORS AND THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

The diminished renal blood flow characteristic of states in which the effective arterial blood volume is reduced is translated by the renal

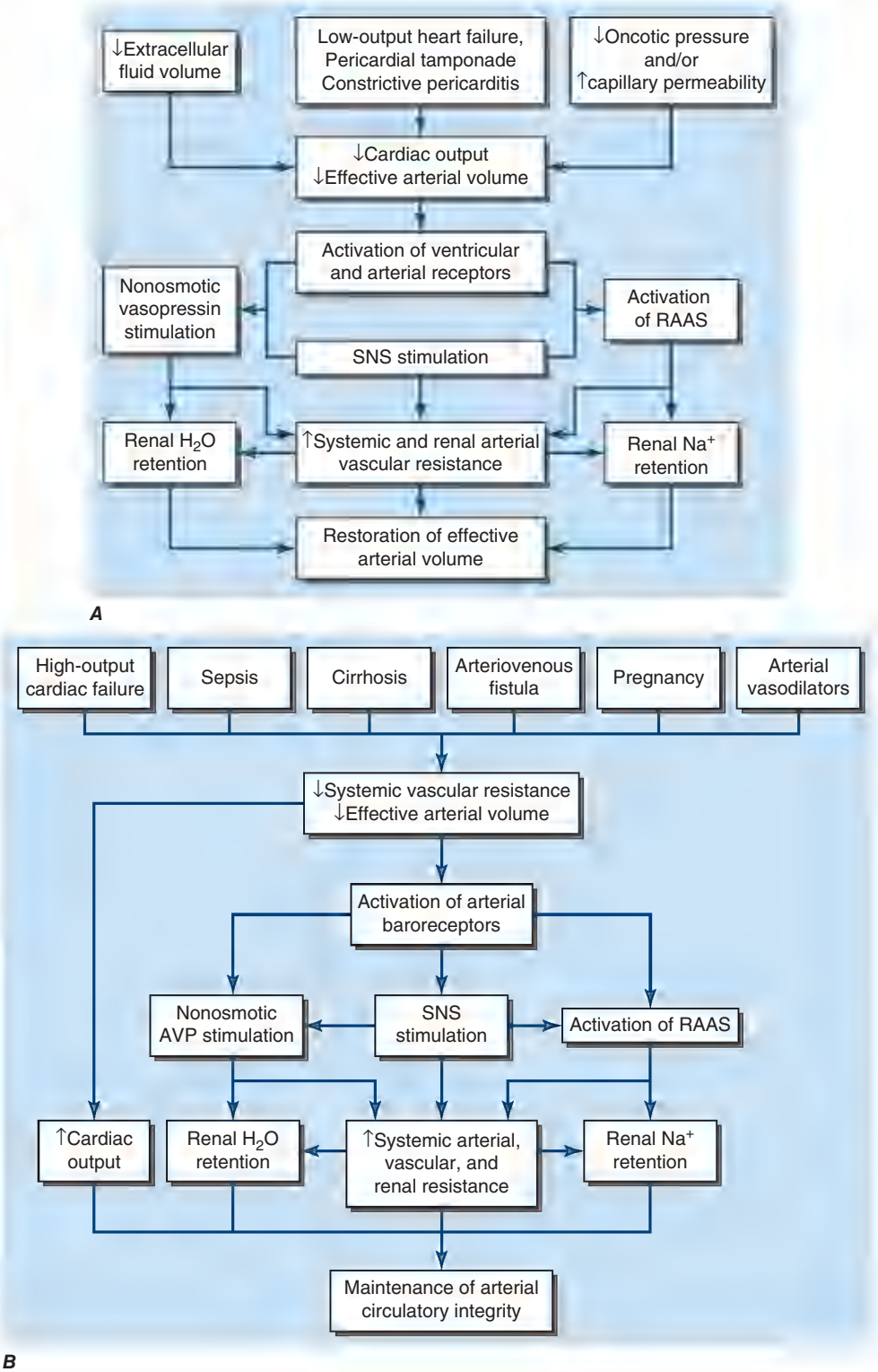

FIGURE 43-1 Clinical conditions in which a decrease in cardiac output (*A*) and systemic vascular resistance (*B*) cause arterial underfilling with resulting neurohumoral activation and renal sodium and water retention. In addition to activating the neurohumoral axis, adrenergic stimulation causes renal vasoconstriction and enhances sodium and fluid transport by the proximal tubule epithelium. AVP, arginine vasopressin; RAAS, renin-angiotensin aldosterone system; SNS, sympathetic nervous system. *(From Annals of Internal Medicine, RW Schrier: Body fluid volume regulation in health and disease: A unifying hypothesis. 113(2):155-159, 1990. Copyright © 1990, American College of Physicians. All Rights Reserved. Reprinted with the permission of American College of Physicians, Inc.)*

juxtaglomerular cells (specialized myoepithelial cells surrounding the afferent arteriole) into a signal for increased renin release. Renin is an enzyme with a molecular mass of about 40,000 Da that acts on its substrate, angiotensinogen, an α_2-globulin synthesized by the liver, to release angiotensin I, a decapeptide, which in turn is converted to angiotensin II (AII), an octapeptide. AII has generalized vasoconstrictor properties, particularly on the renal efferent arterioles. This action reduces the hydrostatic pressure in the peritubular capillaries, whereas the increased filtration fraction raises the colloid osmotic pressure in these vessels, thereby enhancing salt and water reabsorption in the proximal tubule as well as in the ascending limb of the loop of Henle.

The renin-angiotensin-aldosterone system (RAAS) operates as both a hormonal and paracrine system. Its activation causes sodium and water retention and thereby contributes to edema formation. Blockade of the conversion of angiotensin I to AII and blockade of the AII receptors enhance sodium and water excretion and reduce many forms of edema. AII that enters the systemic circulation stimulates the production of aldosterone by the zona glomerulosa of the adrenal cortex. Aldosterone in turn enhances sodium reabsorption (and potassium excretion) by the collecting tubule, further favoring edema formation. Blockade of the action of aldosterone by spironolactone or eplerenone (aldosterone antagonists) or by amiloride (a blocker of epithelial sodium channels) often induces a moderate diuresis in edematous states.

ARGININE VASOPRESSIN

(See also Chap. 393) The secretion of arginine vasopressin (AVP) by the posterior pituitary gland occurs in response to increased intracellular osmolar concentration; by stimulating V_2 receptors, AVP increases the reabsorption of free water in the distal tubules and collecting ducts of the kidney, thereby increasing total body water. Circulating AVP is elevated in many patients with heart failure secondary to a nonosmotic stimulus associated with decreased effective arterial volume and reduced compliance of the left atrium. Such patients fail to show the normal reduction of AVP with a reduction of osmolality, contributing to edema formation and hyponatremia.

ENDOTHELIN-1

This potent peptide vasoconstrictor is released by endothelial cells. Its concentration in the plasma is elevated in patients with severe heart failure and contributes to renal vasoconstriction, sodium retention, and edema.

NATRIURETIC PEPTIDES

Atrial distention causes release into the circulation of atrial natriuretic peptide (ANP), a 3000-Da polypeptide. A high-molecular-weight precursor of ANP is stored in secretory granules within atrial myocytes. A closely related natriuretic peptide (pre-prohormone brain natriuretic peptide [BNP]) is stored primarily in ventricular myocytes and is released when ventricular diastolic pressure rises. Released ANP and BNP (which is derived from its precursor) bind to the natriuretic receptor-A, which

causes (1) excretion of sodium and water by augmenting glomerular filtration, inhibiting sodium reabsorption in the proximal tubule, and inhibiting release of renin and aldosterone; and (2) dilation of arterioles and venules by antagonizing the vasoconstrictor actions of AII, AVP, and sympathetic stimulation. Thus, elevated levels of natriuretic peptides have the capacity to oppose sodium retention in hypervolemic and edematous states.

Although circulating levels of ANP and BNP are elevated in heart failure and in cirrhosis with ascites, these natriuretic peptides are not sufficiently potent to prevent edema formation. Indeed, in edematous states, resistance to the actions of natriuretic peptides may be increased, further reducing their effectiveness.

Further discussion of the control of sodium and water balance is found in **Chap. S1**.

CLINICAL CAUSES OF EDEMA

A weight gain of several kilograms usually precedes overt manifestations of generalized edema. *Anasarca* refers to gross, generalized edema. *Ascites* **(Chap. 53)** and *hydrothorax* refer to accumulation of excess fluid in the peritoneal and pleural cavities, respectively, and are considered special forms of edema.

Edema is recognized by the persistence of an indentation of the skin after pressure, known as "pitting" edema. In its more subtle form, edema may be detected by noting that after the stethoscope is removed from the chest wall, the rim of the bell leaves an indentation on the skin of the chest for a few minutes. Edema may be present when the ring on a finger fits more snugly than in the past or when a patient complains of difficulty putting on shoes, particularly in the evening when dependent edema is greatest. Edema may also be recognized by puffiness of the face (most prominent upon awakening), which is most readily apparent in the periorbital areas owing to relative tissue laxity.

GENERALIZED EDEMA

The differences among the major causes of generalized edema are shown in **Table 43-1**. Cardiac, renal, hepatic, or nutritional disorders are responsible for a large majority of patients with generalized edema. Consequently, the differential diagnosis of generalized edema should be directed toward identifying or excluding these several conditions.

Heart Failure **(See also Chap. 264)** In heart failure, the impaired systolic emptying of the ventricle(s) and/or the impairment of ventricular relaxation promotes an accumulation of blood in the venous circulation at the expense of the effective arterial volume. In addition, activation of the sympathetic nervous system and the RAAS (see above) acts in concert to cause renal vasoconstriction and reduction of glomerular filtration and salt and water retention. Sodium and water retention continue, and the increment in blood volume accumulates in the venous circulation, raising venous and intracapillary pressure and resulting in edema (Fig. 43-1).

The presence of overt cardiac disease, as manifested by cardiac enlargement and/or ventricular hypertrophy, together with clinical evidence of cardiac failure, such as dyspnea, basilar rales, venous distention, and hepatomegaly, usually indicates that edema results from heart failure. Noninvasive tests such as electrocardiography, echocardiography, and measurements of BNP (or N-terminal proBNP [NT-proBNP]) are helpful in establishing the diagnosis of heart disease. The edema of heart failure typically occurs in the dependent portions of the body.

Edema of Renal Disease **(See also Chap. 326)** The edema that occurs during the acute phase of glomerulonephritis is characteristically associated with hematuria, proteinuria, and hypertension. In most instances, the edema results from primary retention of sodium and water by the kidney owing to renal dysfunction. This state differs from most forms of heart failure in that it is characterized by a normal (or sometimes even increased) cardiac output. Patients with *chronic* renal failure may also develop edema due to primary renal retention of sodium and water.

Nephrotic Syndrome and Other Hypoalbuminemic States The primary alteration in the nephrotic syndrome is a diminished colloid oncotic pressure due to losses of large quantities (≥3.5 g/d) of protein into the urine and hypoalbuminemia (<3.0 g/dL). As a result of the reduced colloid osmotic pressure, the sodium and water that are retained cannot be confined within the vascular compartment, and total and effective arterial blood volumes decline. This process initiates the edema-forming sequence of events described above, including activation of the RAAS. The nephrotic syndrome may occur during the course of a variety of kidney diseases, including glomerulonephritis, diabetic glomerulosclerosis, and hypersensitivity reactions. The edema is diffuse, symmetric, and most prominent in the dependent areas; periorbital edema is most prominent in the morning.

Hepatic Cirrhosis **(See also Chap. 355)** This condition is characterized, in part, by hepatic venous outflow obstruction, which in turn expands the splanchnic blood volume, and hepatic lymph formation. Intrahepatic hypertension acts as a stimulus for renal sodium retention

TABLE 43-1 Principal Causes of Generalized Edema: History, Physical Examination, and Laboratory Findings

ORGAN SYSTEM	HISTORY	PHYSICAL EXAMINATION	LABORATORY FINDINGS
Cardiac	Dyspnea with exertion prominent—often associated with orthopnea—or paroxysmal nocturnal dyspnea	Elevated jugular venous pressure, ventricular (S_3) gallop; occasionally with displaced or dyskinetic apical impulse; peripheral cyanosis, cool extremities, low pulse pressure when severe	Elevated urea nitrogen-to-creatinine ratio common; serum sodium often decreased; elevated natriuretic peptides
Hepatic	Dyspnea uncommon, except if associated with significant degree of ascites; most often a history of ethanol abuse	Frequently associated with ascites; jugular venous pressure normal or low; blood pressure lower than in renal or cardiac disease; one or more additional signs of chronic liver disease (jaundice, palmar erythema, Dupuytren's contracture, spider angiomata, male gynecomastia; asterixis and other signs of encephalopathy) may be present	If severe, reductions in serum albumin, cholesterol, other hepatic proteins (transferrin, fibrinogen); liver enzymes elevated, depending on the cause and acuity of liver injury; tendency toward hypokalemia, respiratory alkalosis; macrocytosis from folate deficiency
Renal (CRF)	Usually chronic: may be associated with uremic signs and symptoms, including decreased appetite, altered (metallic or fishy) taste, altered sleep pattern, difficulty concentrating, restless legs, or myoclonus; dyspnea can be present, but generally less prominent than in heart failure	Elevated blood pressure; hypertensive retinopathy; uremic fetor; pericardial friction rub in advanced cases with uremia	Elevation of serum creatinine and cystatin C; albuminuria; hyperkalemia, metabolic acidosis, hyperphosphatemia, hypocalcemia, anemia (usually normocytic)
Renal (NS)	Childhood diabetes mellitus; plasma cell dyscrasias	Periorbital edema; hypertension	Proteinuria (≥3.5 g/d); hypoalbuminemia; hypercholesterolemia; microscopic hematuria

Abbreviations: CRF, chronic renal failure; NS, nephrotic syndrome.

Source: Reproduced with permission from GM Chertow, E Braunwald, L Goldman (eds). Approach to the patient with edema, in Primary Cardiology, 2nd ed. Philadelphia, Saunders, 2003.

and causes a reduction of effective arterial blood volume. These alterations are frequently complicated by hypoalbuminemia secondary to reduced hepatic synthesis, as well as peripheral arterial vasodilation. These effects reduce the effective arterial blood volume, leading to activation of the sodium- and water-retaining mechanisms described above **(Fig. 43-1*B*)**. The concentration of circulating aldosterone often is elevated by the failure of the liver to metabolize this hormone. Initially, the excess interstitial fluid is localized preferentially proximal (upstream) to the congested portal venous system, causing ascites **(Chap. 53)**. In later stages, particularly when there is severe hypoalbuminemia, peripheral edema may develop. A sizable accumulation of ascitic fluid may increase intraabdominal pressure and impede venous return from the lower extremities, thereby contributing to the accumulation of the edema.

Drug-Induced Edema A large number of widely used drugs can cause edema **(Table 43-2)**. Mechanisms include renal vasoconstriction (nonsteroidal anti-inflammatory drugs and cyclosporine), arteriolar dilation (vasodilators), augmented renal sodium reabsorption (steroid hormones), and capillary damage.

Edema of Nutritional Origin A diet grossly deficient in calories and particularly in protein over a prolonged period may produce hypoproteinemia and edema. The latter may be intensified by the development of beriberi heart disease, which also is of nutritional origin, in which multiple peripheral arteriovenous fistulae result in reduced effective systemic perfusion and effective arterial blood volume, thereby enhancing edema formation **(Chap. 344)** (Fig. 43-1*B*). Edema develops or becomes intensified when famished subjects are first provided with an adequate diet. The ingestion of more food may increase the quantity of sodium ingested, which is then retained along with water. So-called refeeding edema also may be linked to increased release of insulin, which directly increases tubular sodium reabsorption. In addition to hypoalbuminemia, hypokalemia and caloric deficits may be involved in the edema of starvation.

LOCALIZED EDEMA

In thrombophlebitis, varicose veins, and primary venous valve failure, the hydrostatic pressure in the capillary bed upstream (proximal) of the obstruction increases so that an abnormal quantity of fluid is transferred from the vascular to the interstitial space, which may give rise to localized edema. The latter may also occur in lymphatic obstruction caused by chronic lymphangitis, resection of regional lymph nodes, filariasis, and genetic (frequently called primary) lymphedema. The latter is particularly intractable because restriction of lymphatic flow results in both an increase in intracapillary pressure and increased protein concentration in the interstitial fluid, which act in concert to aggravate fluid retention.

Other Causes of Edema These causes include hypothyroidism (myxedema) due to tissue deposition of hyaluronic acid; hyperthyroidism (pretibial myxedema secondary to Graves' disease), in which edema is typically nonpitting; hypercortisolism; pregnancy; and administration of estrogens and vasodilators, particularly calcium channel blockers.

TABLE 43-2 Drugs Associated with Edema Formation

Nonsteroidal anti-inflammatory drugs
Antihypertensive agents
Direct arterial/arteriolar vasodilators
Hydralazine
Clonidine
Methyldopa
Guanethidine
Minoxidil
Calcium channel antagonists
α-Adrenergic antagonists
Thiazolidinediones
Steroid hormones
Glucocorticoids
Anabolic steroids
Estrogens
Progestins
Cyclosporine
Growth hormone
Immunotherapies
Interleukin 2
OKT3 monoclonal antibody

Source: Reproduced with permission from GM Chertow, in E Braunwald, L Goldman (eds): Approach to the patient with edema, in Primary Cardiology, 2nd ed. Philadelphia, Saunders, 2003.

DISTRIBUTION OF EDEMA

The distribution of edema is an important guide to its cause. Edema associated with heart failure tends to be more extensive in the legs and to be accentuated in the evening, a feature also determined largely by posture. When patients with heart failure are confined to bed, edema may be most prominent in the presacral region.

Edema resulting from hypoproteinemia, as occurs in the nephrotic syndrome, characteristically is generalized, but it is especially evident in the very soft tissues of the eyelids and face and tends to be most pronounced in the morning owing to the recumbent posture assumed during the night. Less common causes of facial edema include trichinosis, allergic reactions, and myxedema. Edema limited to one leg or to one or both arms is usually the result of venous and/or lymphatic obstruction. Unilateral paralysis reduces lymphatic and venous drainage on the affected side and may also be responsible for unilateral edema. In patients with obstruction of the superior vena cava, edema is confined to the face, neck, and upper extremities in which the venous pressure is elevated compared with that in the lower extremities.

APPROACH TO THE PATIENT

Edema

An important first question is whether the edema is localized or generalized. If it is localized, the local phenomena that may be responsible should be identified. If the edema is generalized, one should determine if there is serious hypoalbuminemia, e.g., serum albumin <3.0 g/dL. If so, the history, physical examination, urinalysis, and other laboratory data will help evaluate the question of cirrhosis, severe malnutrition, or the nephrotic syndrome as the underlying disorder. If hypoalbuminemia is not present, one should determine if there is evidence of heart failure severe enough to promote generalized edema. Finally, it should be ascertained as to whether or not the patient has an adequate urine output or if there is significant oliguria or anuria. **These abnormalities are discussed in Chaps. 55, 321, and 322.**

FURTHER READING

Clark AL, Cleland JG: Causes and treatment of oedema in patients with heart failure. Nature Rev Cardiol 10:156, 2013.

Frison S et al: Omitting edema measurement: How much acute malnutrition are we missing? Am J Clin Nutr 102:1176, 2015.

Koirala A et al: Etiology and management of edema: A review. Adv Kidney Dis Health 30:110, 2023.

Levick JR, Michel CC: Microvascular fluid exchange and the revised Starling principle. Cardiovascular Res 87:198, 2010.

Telinius N, Hjortdal VE: Role of the lymphatic vasculature in cardiovascular medicine. Heart 105:1777, 2019.

44 Approach to the Patient with a Heart Murmur

Patrick T. O'Gara, Joseph Loscalzo

The differential diagnosis of a heart murmur begins with a careful assessment of its major attributes and response to bedside maneuvers. The history, clinical context, and associated physical examination findings provide additional clues to help establish its significance. Accurate bedside identification of a heart murmur can inform decisions regarding the indications for noninvasive testing and the need for referral to a cardiovascular specialist. Preliminary discussions can be held with the patient regarding natural history, antibiotic use for infective endocarditis or rheumatic fever prophylaxis, the need to restrict various forms of physical activity, and the potential role for family screening.

Heart murmurs are caused by audible vibrations that are due to increased turbulence from accelerated blood flow through normal or abnormal orifices; flow through a narrowed or irregular orifice into a dilated vessel or chamber; or backward flow through an incompetent valve, ventricular septal defect, or patent ductus arteriosus. They traditionally are defined by their timing within the cardiac cycle (Fig. 44-1). *Systolic murmurs* begin with or after the first heart sound (S_1) and terminate at or before the component (A_2 or P_2) of the second heart sound (S_2) that corresponds to their site of origin (left or right, respectively).

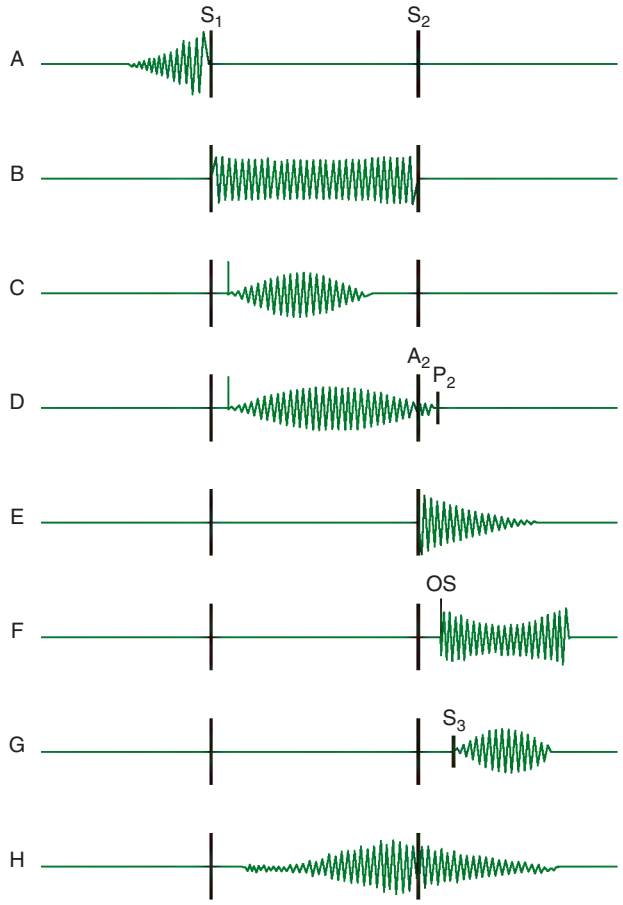

FIGURE 44-1 Diagram depicting principal heart murmurs. *A.* Presystolic murmur of mitral or tricuspid stenosis in a patient with sinus rhythm. ***B.*** Holosystolic (pansystolic) murmur of mitral or tricuspid regurgitation or of ventricular septal defect. ***C.*** Aortic ejection murmur beginning with an ejection click and fading before the second heart sound. ***D.*** Systolic murmur in pulmonic stenosis spilling through the aortic second sound, pulmonic valve closure being delayed. ***E.*** Aortic or pulmonary diastolic murmur. ***F.*** Long diastolic murmur of mitral stenosis after the opening snap (OS). ***G.*** Short mid-diastolic inflow murmur after a third heart sound. ***H.*** Continuous murmur of patent ductus arteriosus. *(Courtesy of Antony and Julie Wood.)*

Diastolic murmurs begin with or after the associated component of S_2 and end at or before the subsequent S_1. *Continuous murmurs* are not confined to either phase of the cardiac cycle but instead begin in early systole and proceed through S_2 into all or part of diastole. The accurate timing of heart murmurs is the first step in their identification. The distinction between S_1 and S_2, and therefore systole and diastole, is usually a straightforward process but can be difficult in the setting of a tachyarrhythmia, in which case the heart sounds can be distinguished by simultaneous palpation of the carotid upstroke, which should closely follow S_1.

Duration and Character The duration of a heart murmur depends on the length of time over which a pressure difference exists between two cardiac chambers, the left ventricle and the aorta, the right ventricle and the pulmonary artery, or the great vessels. The magnitude and variability of this pressure difference, coupled with the geometry and compliance of the involved chambers or vessels, dictate the velocity of flow; the degree of turbulence; and the resulting frequency, configuration, and intensity of the murmur. The diastolic murmur of chronic aortic regurgitation (AR) is a blowing, high-frequency event, whereas the murmur of mitral stenosis (MS), indicative of the left atrial–left ventricular diastolic pressure gradient, is a low-frequency event, heard as a rumbling sound with the bell of the stethoscope. The frequency components of a heart murmur may vary at different sites of auscultation. The coarse systolic murmur of aortic stenosis (AS) may sound higher pitched and more acoustically pure at the apex, a phenomenon eponymously referred to as the *Gallavardin effect*. Some murmurs may have a distinct or unusual quality, such as the "honking" sound appreciated in some patients with mitral regurgitation (MR) due to mitral valve prolapse (MVP).

The configuration of a heart murmur may be described as crescendo, decrescendo, crescendo-decrescendo, or plateau. The decrescendo configuration of the murmur of chronic AR (Fig. 44-1*E*) can be understood in terms of the progressive decline in the diastolic pressure gradient between the aorta and the left ventricle. The crescendo-decrescendo configuration of the murmur of AS reflects the changes in the systolic pressure gradient between the left ventricle and the aorta as ejection occurs, whereas the plateau configuration of the murmur of chronic MR (Fig. 44-1*B*) is consistent with the large and nearly constant pressure difference between the left ventricle and the left atrium.

Intensity The intensity of a heart murmur is graded on a scale of 1–6 (or I–VI). A grade 1 murmur is very soft and is heard only with great effort. A grade 2 murmur is easily heard but not particularly loud. A grade 3 murmur is loud but is not accompanied by a palpable thrill over the site of maximal intensity. A grade 4 murmur is very loud and accompanied by a thrill. A grade 5 murmur is loud enough to be heard with only the edge of the stethoscope touching the chest, whereas a grade 6 murmur is loud enough to be heard with the stethoscope slightly off the chest. Murmurs of grade 3 or greater intensity usually signify important structural heart disease and indicate high blood flow velocity at the site of murmur production. Small, restrictive ventricular septal defects (VSDs), for example, are accompanied by loud, usually grade 4 or greater, systolic murmurs as blood is ejected at high velocity from the left ventricle to the right ventricle through a narrow passage. Low-velocity events, such as left-to-right shunting across an atrial septal defect (ASD), are usually silent. The intensity of a heart murmur may be diminished by any process that increases the distance between the intracardiac source and the stethoscope on the chest wall, such as obesity, obstructive lung disease, or a large pericardial effusion. The intensity of a murmur also may be misleadingly soft when cardiac output is reduced significantly or when the pressure gradient between the involved cardiac structures is low.

Location and Radiation Recognition of the location and radiation of the murmur helps facilitate its accurate identification (Fig. 44-2). Adventitious sounds, such as a systolic click or diastolic snap, or abnormalities of S_1 or S_2 may provide additional clues. Careful attention to the characteristics of the murmur and other heart sounds during the respiratory cycle and the performance of simple bedside maneuvers